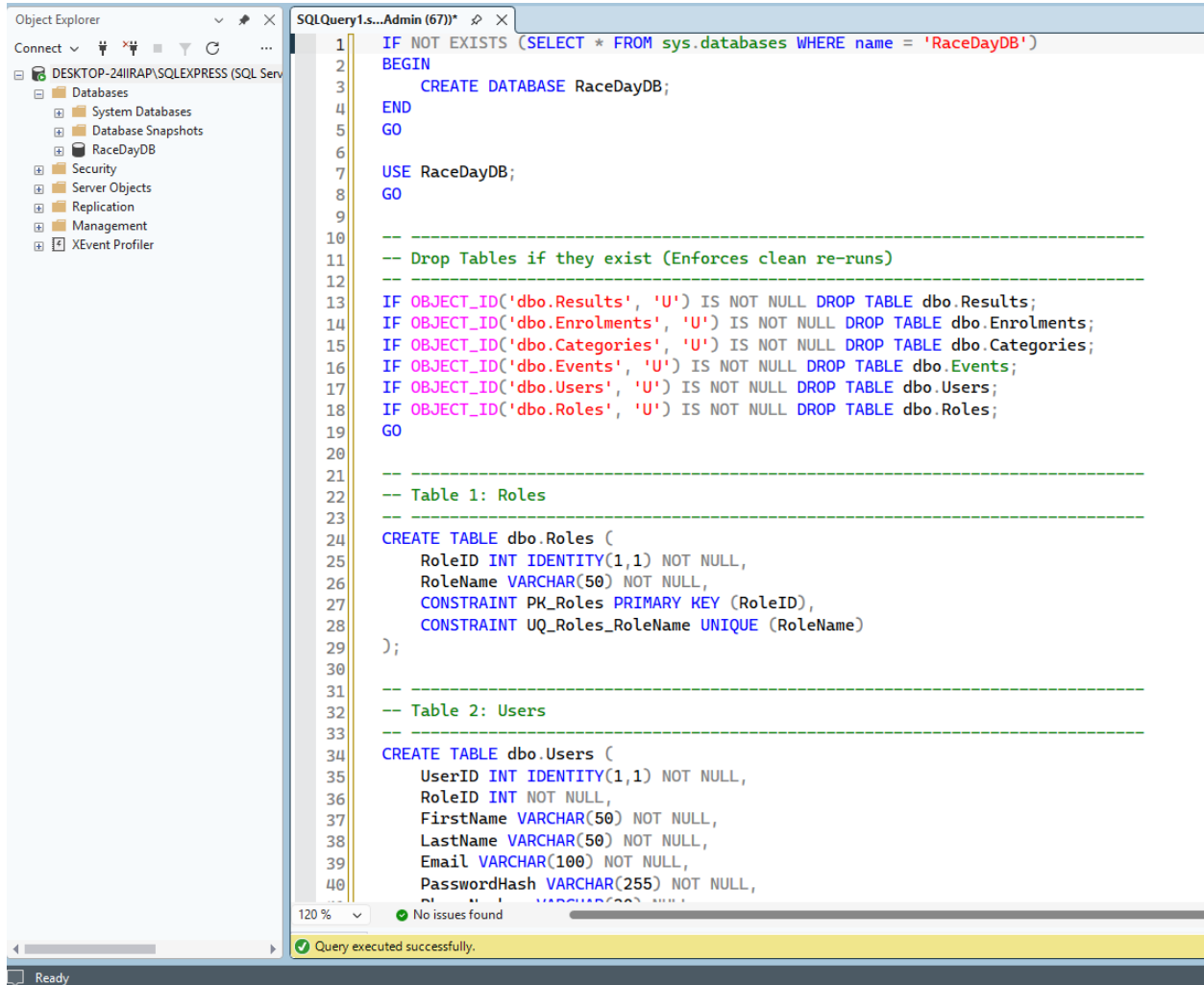


Section: C

SQL Database Script



```
1 IF NOT EXISTS (SELECT * FROM sys.databases WHERE name = 'RaceDayDB')
2 BEGIN
3     CREATE DATABASE RaceDayDB;
4 END
5 GO
6
7 USE RaceDayDB;
8 GO
9
10
11 -- Drop Tables if they exist (Enforces clean re-runs)
12
13 IF OBJECT_ID('dbo.Results', 'U') IS NOT NULL DROP TABLE dbo.Results;
14 IF OBJECT_ID('dbo.Enrolments', 'U') IS NOT NULL DROP TABLE dbo.Enrolments;
15 IF OBJECT_ID('dbo.Categories', 'U') IS NOT NULL DROP TABLE dbo.Categories;
16 IF OBJECT_ID('dbo.Events', 'U') IS NOT NULL DROP TABLE dbo.Events;
17 IF OBJECT_ID('dbo.Users', 'U') IS NOT NULL DROP TABLE dbo.Users;
18 IF OBJECT_ID('dbo.Roles', 'U') IS NOT NULL DROP TABLE dbo.Roles;
19 GO
20
21
22 -- Table 1: Roles
23
24 CREATE TABLE dbo.Roles (
25     RoleID INT IDENTITY(1,1) NOT NULL,
26     RoleName VARCHAR(50) NOT NULL,
27     CONSTRAINT PK_Roles PRIMARY KEY (RoleID),
28     CONSTRAINT UQ_Roles_RoleName UNIQUE (RoleName)
29 );
30
31
32 -- Table 2: Users
33
34 CREATE TABLE dbo.Users (
35     UserID INT IDENTITY(1,1) NOT NULL,
36     RoleID INT NOT NULL,
37     FirstName VARCHAR(50) NOT NULL,
38     LastName VARCHAR(50) NOT NULL,
39     Email VARCHAR(100) NOT NULL,
40     PasswordHash VARCHAR(255) NOT NULL,
```

120 % No issues found

Query executed successfully.

Ready

```

34 CREATE TABLE dbo.Users (
35     UserID INT IDENTITY(1,1) NOT NULL,
36     RoleID INT NOT NULL,
37     FirstName VARCHAR(50) NOT NULL,
38     LastName VARCHAR(50) NOT NULL,
39     Email VARCHAR(100) NOT NULL,
40     PasswordHash VARCHAR(255) NOT NULL,
41     PhoneNumber VARCHAR(20) NULL,
42     CreatedAt DATETIME NOT NULL CONSTRAINT DF_Users_CreatedAt DEFAULT GETDATE(),
43     CONSTRAINT PK_Users PRIMARY KEY (UserID),
44     CONSTRAINT UQ_Users_Email UNIQUE (Email),
45     CONSTRAINT FK_Users_Roles FOREIGN KEY (RoleID) REFERENCES dbo.Roles(RoleID) ON DELETE CASCADE
46 );

```

```

47
48 -----
49 -- Table 3: Events
50 -----

```

```

51 CREATE TABLE dbo.Events (
52     EventID INT IDENTITY(1,1) NOT NULL,
53     OrganiserID INT NOT NULL,
54     EventName VARCHAR(150) NOT NULL,
55     Description VARCHAR(MAX) NULL,
56     Location VARCHAR(150) NOT NULL,
57     EventDate DATETIME NOT NULL,
58     CreatedAt DATETIME NOT NULL CONSTRAINT DF_Events_CreatedAt DEFAULT GETDATE(),
59     CONSTRAINT PK_Events PRIMARY KEY (EventID),
60     CONSTRAINT FK_Events_Organiser FOREIGN KEY (OrganiserID) REFERENCES dbo.Users(UserID)
61 );

```

```

62
63 -----
64 -- Table 4: Categories
65 -----

```

```

66 CREATE TABLE dbo.Categories (
67     CategoryID INT IDENTITY(1,1) NOT NULL,
68     EventID INT NOT NULL,
69     CategoryName VARCHAR(100) NOT NULL,
70     DistanceKM DECIMAL(5,2) NOT NULL,
71     EntryFee DECIMAL(10,2) NOT NULL,
72     MaxParticipants INT NOT NULL,
73     CONSTRAINT PK_Categories PRIMARY KEY (CategoryID),
74     CONSTRAINT FK_Categories_Event FOREIGN KEY (EventID) REFERENCES dbo.Events(EventID) ON DELETE CASCADE

```

120 % No issues found

Query executed successfully.

DES

```
SQLQuery1.s...Admin (677)*
73      CONSTRAINT PK_Categories PRIMARY KEY (CategoryID),
74      CONSTRAINT FK_Categories_Events FOREIGN KEY (EventID) REFERENCES dbo.Events(EventID) ON DELETE CASCADE,
75      CONSTRAINT CK_Categories_Distance CHECK (DistanceKM > 0),
76      CONSTRAINT CK_Categories_Fee CHECK (EntryFee >= 0),
77      CONSTRAINT CK_Categories_MaxPart CHECK (MaxParticipants > 0)
78  );
79
80  -----
81  -- Table 5: Enrolments
82  -----
83  CREATE TABLE dbo.Enrolments (
84      EnrolmentID INT IDENTITY(1,1) NOT NULL,
85      ParticipantID INT NOT NULL,
86      CategoryID INT NOT NULL,
87      EnrolmentDate DATETIME NOT NULL CONSTRAINT DF_Enrolments_EnrolmentDate DEFAULT GETDATE(),
88      PaymentStatus VARCHAR(20) NOT NULL CONSTRAINT DF_Enrolments_PaymentStatus DEFAULT 'Pending',
89      RaceNumber INT NULL,
90      CONSTRAINT PK_Enrolments PRIMARY KEY (EnrolmentID),
91      CONSTRAINT FK_Enrolments_Users FOREIGN KEY (ParticipantID) REFERENCES dbo.Users(UserID),
92      CONSTRAINT FK_Enrolments_Categories FOREIGN KEY (CategoryID) REFERENCES dbo.Categories(CategoryID) ON DELETE CASCADE,
93      CONSTRAINT UQ_Participant_Category UNIQUE (ParticipantID, CategoryID),
94      CONSTRAINT CK_Enrolments_PaymentStatus CHECK (PaymentStatus IN ('Paid', 'Pending', 'Cancelled'))
95  );
96
97  -----
98  -- Table 6: Results
99  -----
100  CREATE TABLE dbo.Results (
101      ResultID INT IDENTITY(1,1) NOT NULL,
102      EnrolmentID INT NOT NULL,
103      FinishTime TIME NULL,
104      OverallPosition INT NULL,
105      CategoryPosition INT NULL,
106      Status VARCHAR(20) NOT NULL CONSTRAINT DF_Results_Status DEFAULT 'Finished',
107      CONSTRAINT PK_Results PRIMARY KEY (ResultID),
108      CONSTRAINT UQ_Results_EnrolmentID UNIQUE (EnrolmentID),
109      CONSTRAINT FK_Results_Enrolments FOREIGN KEY (EnrolmentID) REFERENCES dbo.Enrolments(EnrolmentID) ON DELETE CASCADE,
110      CONSTRAINT CK_Results_Status CHECK (Status IN ('Finished', 'DNF', 'DNS', 'DQ'))
111  );
112  GO
113
114  120 %  No issues found
115  Query executed successfully.  DESKTOP-24IIRAP\SQLEXPRESS ...
```

```
SQLQuery1.s...Admin (67)*
109 CONSTRAINT FK_Results_Enrolments FOREIGN KEY (EnrolmentID) REFERENCES dbo.Enrolments(EnrolmentID) ON DELETE CASCADE,
110 CONSTRAINT CK_Results_Status CHECK (Status IN ('Finished', 'DNF', 'DNS', 'DQ'))
111 );
112 GO
113
114 -- =====
115 -- Seed Sample Data
116 -- =====
117
118 -- 1. Insert Roles
119 INSERT INTO dbo.Roles (RoleName) VALUES
120 ('Organiser'),
121 ('Participant');
122
123 -- 2. Insert Users (2 Organisers, 2 Participants)
124 -- Password hashes represent placeholder hashed values
125 INSERT INTO dbo.Users (RoleID, FirstName, LastName, Email, PasswordHash, PhoneNumber) VALUES
126 (1, 'Sipho', 'Nkosi', 'sipho.nkosi@raceevents.co.za', 'AQAAAAEACcQAAAAEHhashedpass1...', '0821112223'),
127 (1, 'Anika', 'Van Zyl', 'anika@capesports.co.za', 'AQAAAAEACcQAAAAEHhashedpass2...', '0834445556'),
128 (2, 'Thabo', 'Mokoena', 'thabo.m@gmail.com', 'AQAAAAEACcQAAAAEHhashedpass3...', '0717778889'),
129 (2, 'Sarah', 'Jenkins', 'sarah.j@yahoo.com', 'AQAAAAEACcQAAAAEHhashedpass4...', '0790001112');
130
131 -- 3. Insert Events (3 Events)
132 INSERT INTO dbo.Events (OrganiserID, EventName, Description, Location, EventDate) VALUES
133 (1, 'Soweto Marathon 2026', 'The iconic race through historic Soweto landmarks.', 'Soweto, Johannesburg', '2026-11-01 06:00:00'),
134 (2, 'Cape Town Cycle Tour 2027', 'World famous road cycling event around the Cape Peninsula.', 'Cape Town, Western Cape', '2027-03-14 06:30:00'),
135 (1, 'Durban Promenade Fun Run', 'Family-friendly coastal run along the Durban beachfront.', 'Durban, KwaZulu-Natal', '2026-10-15 07:00:00');
136
137 -- 4. Insert Categories
138 INSERT INTO dbo.Categories (EventID, CategoryName, DistanceKM, EntryFee, MaxParticipants) VALUES
139 -- Soweto Marathon Categories
140 (1, '42.2km Full Marathon', 42.20, 350.00, 10000),
141 (1, '21.1km Half Marathon', 21.10, 250.00, 15000),
142 (1, '10km Open Run', 10.00, 150.00, 8000),
143 -- Cape Town Cycle Tour Categories
144 (2, '109km Main Race', 109.00, 650.00, 30000),
145 (2, '42km Short Route', 42.00, 400.00, 5000),
146 -- Durban Promenade Categories
147 (3, '10km Timed Run', 10.00, 120.00, 2000),
148 (3, '5km Fun Walk', 5.00, 80.00, 3000);
149
120 %
No issues found
Query executed successfully.
DESKTOP-24IRAP\SQLEXPRESS ... DESKTOP-24IRAP\Admin ... RaceDayD
```

```
SQLQuery1.s...Admin (67)*
142 (1, '10km Open Run', 10.00, 150.00, 8000),
143 -- Cape Town Cycle Tour Categories
144 (2, '109km Main Race', 109.00, 650.00, 30000),
145 (2, '42km Short Route', 42.00, 400.00, 5000),
146 -- Durban Promenade Categories
147 (3, '10km Timed Run', 10.00, 120.00, 2000),
148 (3, '5km Fun Walk', 5.00, 80.00, 3000);
149
150 -- 5. Insert Enrolments
151 INSERT INTO dbo.Enrolments (ParticipantID, CategoryID, PaymentStatus, RaceNumber) VALUES
152 (3, 1, 'Paid', 1001), -- Thabo enrolled in Soweto Full Marathon
153 (4, 2, 'Paid', 5042), -- Sarah enrolled in Soweto Half Marathon
154 (3, 4, 'Paid', 1209), -- Thabo enrolled in Cape Town Cycle 109km
155 (4, 6, 'Pending', NULL); -- Sarah enrolled in Durban 10km
156
157 -- 6. Insert Results
158 INSERT INTO dbo.Results (EnrolmentID, FinishTime, OverallPosition, CategoryPosition, Status) VALUES
159 (1, '03:15:42', 142, 28, 'Finished'),
160 (2, '01:45:10', 310, 45, 'Finished');
161 GO
162
163
164 USE RaceDayDB;
165 GO
166
167 SELECT * FROM dbo.Roles;
168 SELECT * FROM dbo.Users;
169 SELECT * FROM dbo.Events;
170 SELECT * FROM dbo.Categories;
171 SELECT * FROM dbo.Enrolments;
172 SELECT * FROM dbo.Results;
```

	RoleID	RoleName
1	1	Organiser
2	2	Participant

	UserID	RoleID	FirstName	LastName	Email	PasswordHash	PhoneNumber	CreatedAt
1	1	1	Sipho	Nkosi	sipho.nkosi@raceevents.co.za	AQAAAAEAACQAAAAEHnashedpass1...	0821112223	2026-09-04 15:09:48.087
2	2	1	Anika	Van Zyl	anika@capeports.co.za	AQAAAAEAACQAAAAEHnashedpass2...	0834445555	2026-09-04 15:09:48.087
3	3	2	Thabo	Mokoen	thabo.m@gmail.com	AQAAAAEAACQAAAAEHnashedpass3...	0717778889	2026-09-04 15:09:48.087
4	4	2	Sarah	Jenkins	sarah.j@yahoo.com	AQAAAAEAACQAAAAEHnashedpass4...	0790001112	2026-09-04 15:09:48.087

	EventID	OrganiserID	EventName	Description	Location	EventDate	CreatedAt	
1	1	1	Soweto Marathon 2026	The iconic race through historic Soweto landmarks.	Soweto, Johannesburg	2026-11-01 06:00:00.000	2026-09-04 15:09:48.087	
2	2	2	Cape Town Cycle Tour 2027	World famous road cycling event around the Cape ...	Cape Town, Western Cape	2027-03-14 06:30:00.000	2026-09-04 15:09:48.087	
3	3	1	Durban Promenade Fun Run	Family-friendly coastal run along the Durban beachfr...	Durban, KwaZulu-Natal	2026-10-15 07:00:00.000	2026-09-04 15:09:48.087	

	CategoryID	EventID	CategoryName	DistanceKM	EntryFee	MaxParticipants
1	1	1	42.2km Full Marathon	42.20	350.00	10000
2	2	1	21.1km Half Marathon	21.10	250.00	15000
3	3	1	10km Open Run	10.00	150.00	8000
4	4	2	109km Main Race	109.00	650.00	30000
5	5	2	42km Short Route	42.00	400.00	5000
6	6	3	10km Timed Run	10.00	120.00	2000
7	7	3	5km Fun Walk	5.00	80.00	3000

	EnrolmentID	ParticipantID	CategoryID	EnrolmentDate	PaymentStatus	RaceNumber
1	1	3	1	2026-09-04 15:09:48.087	Paid	1001
2	2	4	2	2026-09-04 15:09:48.087	Paid	5042
3	3	3	4	2026-09-04 15:09:48.087	Paid	1209
4	4	4	6	2026-09-04 15:09:48.087	Pending	NULL

	ResultID	EnrolmentID	FinishTime	OverallPosition	CategoryPosition	Status
1	1	1	03:15:42.0000000	142	28	Finished
2	2	2	01:45:10.0000000	310	45	Finished